

Letters of Euler
ON NATURAL PHILOSOPHY

Transcribed by: Uncle Jeffy
for a Canadian Princess and Prince

Dearest Juliet and Jackson,

Continuation of the Subject, and of Mistakes in
the Knowledge of Truth.

The three classes of truths which I have now
un-folded are the only sources of all our knowledge; all being
derived from our own experience, from reasoning, or from
the report of others.

It is not easy to determine which of these three
sources contributes most to the increase of knowledge. Adam
and Eve must have derived theirs chiefly from the first two;
God, however, revealed many things to them, the knowledge of
which is to be referred to the third source, as neither
their own experience nor their powers of reasoning could
have conducted them so far.

Without recurring to a period so remote, we are sufficiently convinced, that if we were determined to believe nothing of what we hear from others, or read in their writings, we should be in a state of almost total ignorance. It is very far, however, from being our duty to believe every thing that is said, or that we read.

We ought constantly to employ our discerning faculties, not only with respect to truths of the third class, but likewise of the others.

We are so liable to suffer ourselves to be dazzled by the senses, and to mistake in our reasonings, that the very sources laid open by the Creator for the discovery of truth very frequently plunge us into error. Notions of the third class, therefore, ought not in reason to fall under suspicion, any more than such as belong to the other two. We ought, therefore, to be equally on our ~~ground~~ guard against deception, whatever be the class to which the notion belongs; for we find as many instances of error in the first and second classes as in the third.

The same holds with regard to the certainty of the particular articles of knowledge which these three sources supply; and it cannot be affirmed that the truths of any one order have a surer foundation than those of another. Each class is liable to errors, by which we may be misled; but there are likewise precautions which, carefully observed, furnish us with nearly the same degree of conviction.

I do not know whether you are more thoroughly convinced of this truth, that ~~two~~ triangles which have the same base and the same height are equal to one another, than of this, that the Russians have been at Berlin; though the former is founded on a chain of accurate reasoning, whereas the latter depends entirely on the veracity of your informer.

Respecting the truths, therefore, of each of these classes, we must rest satisfied with such proofs as correspond to their nature; and it would be ridiculous to insist upon a geometrical demonstration of the truths of experience, or of history. This is usually the fault

of those who make a bad use of their penetration in intellectual truths, to require mathematical demonstration in proof of all the truths of religion, a great part of which belongs to the third class.

There are persons determined to believe and admit nothing but what they see and touch; whatever you would prove to them by reasoning, be it ever so solid, they are disposed to suspect, unless you place it before their eyes. Chemists, anatomists, and natural philosophers, who employ themselves wholly in making experiments, are most chargeable with this fault. Every thing that the one cannot melt in his crucible, or the other dissect with his scalpel, they reject as unfounded. To no purpose would you speak to them of the qualities and nature of the soul; they admit nothing but what strikes the senses.

Thus, the particular kind of study to which every one is addicted has such a powerful influence on his manner of thinking, that the natural philosopher and chemist will have nothing but experiments, and the geometrician and logician nothing but arguments; which constitute, however, proofs entirely different, the one attached to

the first class, the other to the second, which ought always to be carefully distinguished, according to the nature of ^{the} objects.

But can it be possible that persons should exist who, wholly absorbed in pursuits pertaining to the third class, call only for proofs derived from that source?

I have known some of this description, who, totally devoted to the study of history and antiquity, who admit nothing as true but what you could prove by history, or the authority of some ancient author. They perfectly agree with you respecting the truth of the propositions of Euclid, but merely on the authority of that author, without paying any attention to the demonstrations by which he supports them; they even imagine that the contrary of these propositions might be true, if the ancient geometers had thought proper to maintain it.

This is a source of error which retards many in the pursuit of truth; but we find it rather

among the learned, than among those who are beginning to apply themselves to the study of the sciences. We ought to have no predilection in favour of any of the three species of proofs which each class requires; and provided they are sufficient in their kind, we are bound to admit them.

I have seen or felt, is the proof of the first class. I can demonstrate it, is that of the second: we likewise say, I know it is so. Finally, I receive it on the testimony of persons worthy of credit, or I believe it on solid grounds, is the proof of the third class

4th April, 1761.

~~Notes~~
Notes from Uncle Jeffy:

Honor your mother and father; This is the first and foremost important thing to do when getting out there into the world. I am going to bring these letters to you

because I love you. Uncle Jeffy reads, writes,
and prays. These are ways to learn! It's important
to keep learning! When I shadowed a neurosurgeon,
he taught me that everyone is smart! Your
grandpa taught me to reduce noise in my
life so I can recognize the greatness in
others; that was wise and great. We love you,
and know that you will be great.

So, this first letter taught us that
the three classes of ~~truths~~ come from:

- our own experience
- our reasoning (as a family)
- others

This is the first letter of many to come. I have
chosen Euler's Letters from years of experience,
and will trust this other to teach us:

Natural Philosophy.

Love,

Jefferson